

Immune Part 2: complete study guide

For Exam 1, Tue Oct 13, 2026 (covers Classes 1-5).

Study aid only. For understanding the material, not submitted work.

Sources, in priority order: (1) her slides: the recorded-lecture deck [S#] and the in-class deck [class S#]; (2) her spoken points from the recorded lecture [lecture ~mm:ss at 1x]; (3) your full study sheet, immune 2 pages [sheet p.#]; (4) the vocab list; (5) the study-guide gaps notes. **[textbook — not in lecture]** = background that isn't in her material. ★ **Test cue** = she flagged it for the test.

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Appendix · Filled class study guide 5 (Immune part 2)

1. What she flagged for the test

★ **Test cue** (in lecture order):

- **~0:39:00 — Don't memorize labs per disease:** "you don't need to try and memorize for each separate autoimmune disease... is the CRP up? Is ESR elevated in this disease?" → just ask **"is there inflammation present?"** If yes → inflammatory markers ↑ [lecture ~39:19]
- **~0:39:49 — Only 3 inflammatory markers for the test:** "there are more, but those are the only three I'm going to expect you to recognize for the test" = **CRP, fibrinogen, ESR** [lecture ~39:49]
- **~0:41:23 — Only these autoantibodies for the test:** "these are the only ones that I'll expect you to recognize on the test" = **ANA, anti-DNA, anti-CCP, anti-phospholipids, RF** [S34; lecture ~41:23]
- **~0:51:58 — Corticosteroids:** "for now, I want you to know... corticosteroids are powerful anti-inflammatory drugs, but they do have a lot of side effects" [S44; lecture ~51:58]
- **~0:57:40 — Chemical mediators slide:** "you don't have to memorize this slide" (histamine, prostaglandins, leukotrienes, bradykinin, cytokines) [lecture ~57:40]
- **~1:36:11 — Classification cue:** "on the test, I cannot describe a patient's symptoms to you and expect you to be able to tell me, is this a type 2 reaction? Is this a type 4 reaction?" → **she will NOT ask you to classify a described patient into Type II vs IV etc.** [lecture ~1:36:11]
- **★ Exception to that cue (~1:36:52):** if a described scenario **sounds like an allergy** → **answer is Type I, IgE-mediated** [lecture ~1:36:52]
- **~1:37:23 — "the last thing I want you to learn from the slide":** immediate vs delayed timing — Types I-III immediate, Type IV delayed [lecture ~1:37:23]

Also test-relevant (class practice questions, not walked through in the recording): - Which lab most reliably indicates **infection**? (positive urine culture vs $\uparrow$ ESR/ $\uparrow$ WBC/ANA) [class S8] - Highest risk for autoimmune disease: **female whose mother has SLE** [class S9] - Short answer: why must T cells see antigen on the **correct class of MHC**? [class S10] - Short answer: why might **group A beta-hemolytic strep** predispose to autoimmune disease? [class S11]

2. Self vs non-self: MHC and self-tolerance

Goal: self-tolerance

- **Self-tolerance** = immune system tolerates self-cells; no immune response against own cells/antigens [lecture ~01:04]
- Failure $\rightarrow$ immune system attacks a self-antigen $\rightarrow$ **autoimmune disease** [lecture ~01:37]
- **Self-reactive** = white blood cells that "might go out and attack self cells" [lecture ~02:13]

Three mechanisms of self-tolerance

- **1. Regulatory (suppressor) T cells** — 2 jobs: shut off the immune response after the pathogen is gone + **suppress potentially self-reactive WBCs** and get rid of them [lecture ~02:13-02:48; sheet p.17]
- **2. Bone marrow & thymus screening** — B cells mature in bone marrow, T cells in thymus; while immature, these organs screen for self-reactive cells and **destroy them before release into circulation** [lecture ~02:48-03:20]
- **3. MHC molecules** — "the most important way" [lecture ~03:20]

MHC basics

- **MHC = major histocompatibility complex**; aka "self-recognition proteins" [S7; lecture ~03:51]
- Found on the surface of **almost every cell** [lecture ~03:51]
- **Role: display antigens for T cells to inspect** [S7; lecture ~03:51]
- No foreign antigen displayed $\rightarrow$ T cell does nothing $\rightarrow$ normal cells not attacked [lecture ~05:00]

Class I vs Class II MHC

Feature	Class I MHC	Class II MHC
Found on	All nucleated body cells (almost all other cells)	APCs / phagocytic cells [S9; lecture ~06:04-06:34]
Antigens displayed	Only antigens inside the cell : self, abnormal/cancer , mostly viral	Anything the cell phagocytized (many foreign antigens) [lecture ~09:18-10:25]
Works with	Cytotoxic T cells (CD8+)	Helper T cells (CD4+) [S9; lecture ~10:55]
If it finds its antigen	Cytotoxic T releases cytokines $\rightarrow$ destroys that cell	Helper T activates $\rightarrow$ B cells + other T cells $\rightarrow$ adaptive immunity [lecture ~11:29; ~07:41]

- **Memory trick:** $CD4 \times 2 = 8 \rightarrow$ helper T + class II; $CD8 \times 1 = 8 \rightarrow$ cytotoxic T + class I [lecture ~14:44]
- **Protective rule (know this concept):** the T cell must see its **specific foreign antigen displayed on the correct class of MHC** or **it will do nothing** $\rightarrow$ prevents accidental activation and autoimmune attack [lecture ~08:12, ~11:29; class S10]
- Class I is limited because those cells can't phagocytose $\rightarrow$ viral antigens most likely (viruses must enter the cell); bacteria usually stay outside [lecture ~09:49-10:25]

MHC genes, HLA, tissue typing

- MHC is huge $\rightarrow$ coded by **128 different genes**, each with **50-650 alleles** [lecture ~16:58, ~17:29; sheet p.17]
- $\rightarrow$ virtually impossible for 2 people to have the same MHC genotype **unless identical twins** [lecture ~18:00]
- *Textbook note:* full siblings have about a 25% chance of being an HLA match. [textbook — not in lecture]

- Genotype differences do **not** affect how well MHC works (eye-color analogy) [lecture ~18:34]
- **Tissue typing for transplants:** compare MHC genes to see if donated tissue (e.g., kidney) is close enough to avoid rejection [lecture ~19:08]
- **Forensic exams:** tissue under a murder victim's fingernails matched to a suspect [lecture ~19:40]
- **Paternity evaluations** [lecture ~20:10]
- **HLA = Human Leukocyte Antigen** = older term for MHC; "HLA testing" in the news = looking at the DNA that codes for MHC molecules [lecture ~20:41]
- *Textbook note:* HLA is still the current name for the human MHC. [textbook — not in lecture]

Sequestered antigens

- **Sequestered = hidden away / isolated** (like a sequestered jury) [lecture ~27:52]
- Some self-antigens are hidden from the immune system during embryonic/fetal life → the immune system **never learns they are self** [lecture ~28:25]
- Later, tissue damage exposes them → immune system calls them foreign → autoimmune attack [lecture ~28:58; sheet p.18]

3. Autoimmunity

Definition

- **Auto = self** (automatic, automobile) [lecture ~21:45]
- **Autoimmunity = the immune system attacks self-cells/self-antigens** → self-tissue injury → organ/body system doesn't work properly [lecture ~21:45-22:17]
- **Autoantibody = an antibody attacking self-antigens;** expected in autoimmune disease [lecture ~40:21]
- We **don't really know why** autoimmune disease happens — theories only [lecture ~22:52]

Theories of development

- **Damaged self-antigen structure** — infection, inflammation, or a drug/chemical changes the self-antigen so it's not recognized [lecture ~22:52-23:27; sheet p.18]
- **Faulty MHC/T-cell interaction** [lecture ~23:58]
- **Disorder of immune surveillance** — bone marrow/thymus fail to screen out self-reactive cells; or **regulatory/suppressor T cells** not doing their job [lecture ~24:31-25:04]
- **Antigenic mimicry** — foreign antigen looks like self-antigen; antibodies made against it attack self [lecture ~25:39]
- **Group A beta-hemolytic strep** antigen resembles heart & kidney antigens → antibodies → **rheumatic heart disease** or **glomerulonephritis** [lecture ~26:11-26:43]
 - *Textbook note:* rheumatic heart disease is the classic mimicry example; post-strep glomerulonephritis is usually immune-complex (Type III). [textbook — not in lecture]
- Why doctors culture strep throats and give antibiotics — kill it before antibodies form [lecture ~27:18]
- **Release of sequestered antigens** [lecture ~27:52-28:58]

Predisposing factors (S24-S27)

- **Genetic factors** — autoimmune diseases **run in families**; more common with certain **MHC/HLA genotypes** [S25; lecture ~30:00]
- **"Trigger" events** — genetics alone aren't enough; triggers maybe: virus/pathogen damaging a self-antigen, injury exposing a sequestered antigen, chemical binding to a self-antigen [S26; lecture ~30:34-31:04]
- **Gender** — more common in **women**; thought that **estrogen stimulates** the immune system, **androgens suppress** it (women → more autoimmune disease, men → more infections) [S27; lecture ~31:37-32:07]
- Diagnosed/worsened during hormonal shifts: **puberty, pregnancy** [lecture ~32:07]

Remissions / exacerbations (flares)

- Symptoms vary in intensity over time [S28; lecture ~32:40]
- Exacerbation / flare** = symptoms get worse; **remission** = symptoms improve [lecture ~32:40]
- Most autoimmune diseases are **chronic, cycles of remissions and exacerbations** — a few are acute and go away [lecture ~34:55]
- Goal of care: minimize exacerbations, get patients into remission, keep them there as long as possible** [lecture ~33:14]

Organ-specific vs systemic

Type	Meaning	Examples she names
Organ-specific	Damage limited to one organ/tissue	Type 1 diabetes (beta cells, endocrine pancreas); thyroid diseases (autoimmune in origin); ulcerative colitis (intestines) [S29; lecture ~33:34-34:05]
Systemic	Affects tissues in many body systems; targets connective tissue / collagen	Rheumatoid arthritis, lupus [S29; lecture ~34:24]

- Assessment findings vary by which tissue is attacked, severity, and acute vs chronic [lecture ~34:55]

Diagnostic tests (exactly as on her slides)

Inflammatory markers (S30-S34)	Auto-antibodies (S34)
C-reactive protein (CRP)	Anti-nuclear antibodies (ANA)
Fibrinogen	Anti-DNA
Erythrocyte sedimentation rate (ESR)	Anti-CCP
	Anti-phospholipids
	Rheumatoid factor (RF)

- Two primary groups of tests: inflammatory markers + autoantibodies** [S30; lecture ~35:28]
- Inflammatory markers** = markers found in blood in the presence of inflammation; inflammation = tissue's response to injury → elevated markers mean **injury somewhere** (not where, not why) [lecture ~35:28-36:00]
- Autoimmune disease attacks body tissue → injury → markers expected to be elevated [lecture ~36:00]
- CRP** — made mainly in the **liver**; regulates release of **cytokines** → regulates inflammation [S31; lecture ~36:33; sheet p.17]
- Fibrinogen** — yes it's a clotting factor, but also ↑ in inflammation; helps **locate pathogens** so they can be destroyed [S32; lecture ~37:04]
- ESR (sed rate)** — spin blood in centrifuge, time how fast RBCs sediment; **inflammatory markers stick to RBCs → heavier → drop faster → ESR ↑ with inflammation** [S33; lecture ~37:37-38:43]
- Textbook note:* the standard ESR lets blood settle by gravity in an upright tube for 1 hr. Fibrinogen and other proteins make RBCs clump (rouleaux), so they fall faster. On her test, use her explanation. **[textbook — not in lecture]**
- ★ Test cue: only these **three** markers to recognize [lecture ~39:49]; don't memorize which is up in which disease — ask "is there inflammation?" [lecture ~39:19]
- Autoantibodies:** not every autoimmune disease tests positive for the same ones; some raise only one, some several; a positive result **doesn't tell you which** disease [lecture ~40:21-41:23]
- All three problem types raise inflammatory markers** (immunodeficiency, hypersensitivity, autoimmunity) [class S5]
- How to tell them apart:** immunodeficiency → infection (cultures, WBC); hypersensitivity → allergic reactions (allergy testing); **autoimmunity → damage to self tissue → check for autoantibodies** [class S4-S5]

Targets of intervention (S35)

- **1. Stop the underlying mechanism** — suppress the immune system so it stops attacking self: **immunosuppressive drugs** or **plasmapheresis** [S35; lecture ~41:58]
- Slide 44 list: **immunomodulators, immunosuppressive drugs, monoclonal antibodies, corticosteroids** [S44]
- **2. Treat the symptoms** — depends on the disease [lecture ~42:29]
- Type 1 diabetes → **insulin injections** [lecture ~42:34]
- Rheumatoid arthritis → **pain medication / anti-inflammatory** for swollen joints [lecture ~42:44]
- **Anti-inflammatory drugs: NSAIDs** (Motrin/ibuprofen/aspirin) and **corticosteroids** [S44]

4. Systemic lupus erythematosus (SLE)

Definition & mechanism

- **SLE = systemic lupus erythematosus**; "lupus"; one of the more common autoimmune diseases [S36; lecture ~43:12]
- Immune system makes antibodies against **a variety of self-antigens** [lecture ~43:32]
- **ANA (anti-nuclear antibodies)** = a primary autoantibody; **antibodies that attack antigens in the nucleus** [lecture ~43:32; sheet p.17]
- Process: cells damaged → **DNA in nucleus exposed** → immune system makes ANA → ANA binds DNA antigen on other injured cells → **antigen-antibody (immune) complexes** [lecture ~44:02-44:33]
- Not the complex itself → complexes **deposit in tissue** → **activate complement** → **inflammation** = what causes the symptoms [lecture ~44:33-45:04]
- Lupus = a **Type III, immune complex-mediated** hypersensitivity example [lecture ~45:04; ~1:34:19]
- Other autoantibodies made against **RBCs, platelets**, other antigens [lecture ~45:39]

Who gets it

- Autoimmune diseases generally: **more common in women** (estrogen), often **run in families**, diagnosed/worsen at **puberty and pregnancy** [lecture ~31:37-32:07]
- ★ She does **not** give age/sex/ethnicity statistics specific to lupus in this lecture [lecture ~31:37]

Why so many organs

- Antigen-antibody complexes deposit in **tissues with lots of collagen**; collagen is everywhere → **symptoms in many body systems** [lecture ~46:14; sheet p.17]

Assessment findings

- **Skin: bruising, rashes** (skin has lots of collagen) [lecture ~46:46]
- **Malar / butterfly rash** — classic; butterfly shape: **wings on the cheeks, body over the bridge of the nose** [S40; lecture ~46:46-47:19]
- **Sun exposure worsens** skin signs and **frequently triggers exacerbations** → keep lupus patients **out of the sun** [lecture ~47:19]
- **Joints** — lupus arthritis [lecture ~47:50]
- **Kidney** — inflammation, maybe **renal failure** [lecture ~47:50]
- **Lungs, heart, eyes, GI tract** [lecture ~47:50; sheet p.17]
- **Blood cells** — RBCs, platelets attacked [lecture ~45:39; sheet p.17]
- Doesn't hit every system in every patient → **10 patients can present 10 different ways** [lecture ~48:21]

Tests & diagnosis

- **Inflammatory markers** (CRP, fibrinogen, ESR) ↑ — tells you **inflammation is present**, not the cause [S37/S43; lecture ~49:55]
- **Autoantibodies present** — ANA primary, plus anti-DNA, anti-CCP, anti-phospholipids, RF on the slide list — tells you **"probably autoimmune," not which**

one [S43; lecture ~41:23; ~49:55]

- **No single definitive test** for lupus; "the great imitator" [lecture ~48:21-48:53]
- **Diagnosis of exclusion** — rule out other diseases (kidney disease, other arthritis), look for typical symptoms, remission/exacerbation cycles, elevated inflammatory markers, autoantibodies present [lecture ~49:24-50:27]
- Can take patients **years** to get a diagnosis [lecture ~49:24]

Interventions (S44)

- **Two levels** [lecture ~50:58]
- **Immunomodulators** — "modulate/change immune activity"; goal = **suppress the immune system's attack on self-antigens**: immunosuppressive drugs, monoclonal antibodies, corticosteroids [S44; lecture ~50:58; sheet p.17]
- **Plasmapheresis** also named as a way to stop the disease process where the immune system attacks self (general autoimmune intervention) [lecture ~41:58]
- **Anti-inflammatory drugs** for symptoms/inflammation [S44; lecture ~50:58]
- **Corticosteroids** — synthetic cortisol; **more powerful**, but **lots of side effects** in large amounts [lecture ~51:28-51:58]
- **NSAIDs** (Motrin/ibuprofen/aspirin) — **fewer side effects, less powerful**; enough for arthritis pain but **not enough to halt kidney damage** [lecture ~51:58-52:31]
- **Plan: use NSAIDs first, add corticosteroids if necessary** [lecture ~52:31]

5. Inflammation

Definition

- **Inflammation = the tissue's response to injury** [S45; lecture ~53:07; sheet p.18]
- One of the **4 functions of the immune system** (with immunity, self vs non-self, cancer surveillance) [S2; lecture ~52:31]

Inflammation vs infection (she stresses this)

- **Inflammation** = tissue is injured; the cause can be many things [S46; lecture ~53:37]
- **Infection** = injury caused by the presence of a **pathogen** — pathogen must be present to qualify [lecture ~54:09]
- **Injury WITHOUT infection** (her examples): knife cut, **burn, frostbite**, acid spill, **no blood flow** to fingers [lecture ~53:37]
- **"-itis" = inflammation of** (appendicitis, pericarditis) — NOT automatically infection [lecture ~54:41]

★ Test cue: "-itis" means inflammation; inflammation can be caused by infection but is not always [lecture ~55:13]

Purpose of inflammation (3 things)

- **Limit the spread** of harmful agents [S47; lecture ~55:13]
- **Neutralize / destroy** them [S47; lecture ~55:13]
- **Prepare damaged tissue for repair** [S47; lecture ~55:13]
- These are "really great things" — people just dislike how it feels [lecture ~55:13]

Five cardinal signs — and the mechanism of each

Sign	Mechanism
Redness	Vasodilation → more blood to area [lecture ~58:27; sheet p.18]
Heat	More blood = warmer [lecture ~58:59; sheet p.18]
Swelling (edema)	↑ capillary permeability → fluid leaks into tissues [lecture ~59:31; sheet p.18]
Pain	Goes with swelling [lecture ~1:00:02; sheet p.18]
Loss of function	Swollen parts don't work well (sprained finger can't bend) [lecture ~1:00:02; sheet p.18]

- Her full list: **heat, redness, swelling, pain, loss of function** [lecture ~55:44]

- Loss of function is hopefully **temporary** — but chronic inflammation can make it more **permanent** [lecture ~55:44]

Acute vs chronic (only as she covers it)

- She contrasts **temporary loss of function** (acute) vs inflammation that becomes **chronic → more permanent loss of function** [lecture ~55:44]
- **Chronic irritation/inflammation → metaplasia** (cells change type to tolerate it) [lecture ~1:12:34; sheet p.18]
- Chronic inflammation → **scar tissue forms** → deformity + loss of function [lecture ~1:12:34; sheet p.18]
- Example: **rheumatoid arthritis** joints pulled into deformed positions [lecture ~1:13:09]
- Example: scar tissue in **kidneys** — scar tissue can't filter blood/make urine → lost function [lecture ~1:13:41]
- ⚠️ A formal acute-vs-chronic table (timing, infiltrating cells, healing by scarring vs resolution) is **not in her lecture** [textbook — not in lecture]

Trigger → mast cells / basophils / degranulation

- **Trigger = tissue injury → mast cells and basophils degranulate** and release chemical mediators of inflammation [lecture ~56:48; sheet p.18]
- **Basophil** = granulocyte; granules hold chemical mediators [lecture ~56:48]
- **Mast cell** = "basophil that has moved into the tissues" [lecture ~57:22]
- *Textbook note:* mast cells and basophils are related but separate cell lines; basophils don't turn into mast cells. [textbook — not in lecture]
- **Degranulation** = granules dump their chemicals [lecture ~57:22; sheet p.19]
- Mediators named: **histamine, prostaglandins, leukotrienes, bradykinin, cytokines** [lecture ~57:22; sheet p.18]
- People always think of **histamine** only — there are a whole bunch of others [lecture ~57:36]

★ Test cue: "You don't have to memorize this slide" (the chemical-mediators slide)
[lecture ~57:36]

3 types of responses involved

- **Vascular response, cellular response, acute-phase response**
[S52; lecture ~57:55; sheet p.18]

1. Vascular response (blood vessels; local)

- **Vasodilation** → brings more **blood, O₂, fuel, antibodies** → causes **redness + warmth** [lecture ~58:27; sheet p.18]
- **↑ capillary permeability** → capillaries get "leaky" → fluid escapes → brings **nutrients + antibodies** → causes **edema, pain, loss of function** [lecture ~59:31; sheet p.18]
- Benefits of the leaked fluid (**exudate**): brings nutrients/oxygen/antibodies to prevent infection and promote healing; helps **clean a dirty wound** and **dilute irritants/toxins**; less/stagnant blood flow = contaminants **less likely to spread** to other body parts [lecture ~1:02:13]

2. Cellular response (WBCs; local)

- **Phagocytes migrate to the injury: neutrophils first, then monocyte-macrophages** [lecture ~1:03:53; sheet p.18]
- They **phagocytize** pathogens + dead tissue → prep for repair [lecture ~1:03:53]
- If needed, they act as **APCs** → signal **adaptive immunity** (T cells) so the injury doesn't get infected [lecture ~1:04:24; sheet p.18]

3. Acute-phase response (systemic — see Section 6)

- Fever, leukocytosis, ↑ inflammatory markers, skeletal muscle catabolism
[S55-S57; lecture ~1:04:55]

Exudate types (visible in open wounds)

- **Exudate** = fluid collecting in a wound from ↑ capillary permeability [lecture ~1:00:33]
- **Serous** = clear, light yellow = serum without cells [S53; lecture ~1:01:08; sheet p.18]
- **Serosanguineous** = "sero" + "sanguineous (blood)" = watery blood / pink fluid [S53; lecture ~1:01:08; sheet p.18]
- **Purulent** = **pus** = infected, dead WBCs in it [S53; lecture ~1:01:40; sheet p.18]
- Most of the time exudate is **good**, not gross [lecture ~1:02:13]
- In closed injuries you can't see it, but the same fluid shift happens [lecture ~1:03:22]

★ Test cue: 3 exudate types — serous / serosanguineous / purulent [S53]

6. Acute-phase response and labs

What it is

- The **third response** to chemical mediators; the **systemic** effects of inflammation (the vascular + cellular responses are local) [lecture ~1:04:55; S54]
- Components: **fever, leukocytosis, ↑ inflammatory markers, varying degrees of skeletal muscle catabolism** [S55-S57; sheet p.18]

Fever

- Beneficial effects: **bacteria can't survive** at higher temperature; **white blood cells become more active** [lecture ~1:05:26; sheet p.18]
- Part of inflammation — **not always infection** [lecture ~1:05:58]

Leukocytosis

- **↑ number of WBCs** — not just the ones migrating to the injury, an actual increase in total WBC count [S56; lecture ~1:05:58]
- Clinically we almost always think **infection** when we see fever + ↑ WBC — but it's **inflammation**, and inflammation ≠ always infection [lecture ~1:05:58]
- **Her non-infection examples:**
- Post-op **major back surgery** (6 a.m. surgery; 5 p.m. temp + WBC a bit up) — probably too early for wound infection; it's the **incision = injury** [lecture ~1:07:04]
- **Heart attack** patient — heart muscle died/injured → inflammation → mild fever + ↑ WBC [lecture ~1:08:41]
- **How to tell inflammation vs infection: monitor the trends** (infection → fever/WBC keep climbing; surgical inflammation → peaks then comes down); look for **pus**; do a **wound culture** [lecture ~1:08:11]

★ Test cue: fever + elevated WBC ≠ automatic infection — think "has this patient been injured?" [lecture ~1:07:04]

Skeletal muscle catabolism

- **Catabolism** = breaking down protein → skeletal muscle catabolism = breaking down muscle [lecture ~1:10:21]
- Why: muscle protein → **amino acids** → used to **make antibodies** or **repair damaged tissue** [lecture ~1:10:52]
- Body only mobilizes **the degree of inflammation that's useful**: mosquito bite = redness/swelling/itch, no fever, no muscle breakdown; **appendicitis** = fever + muscle catabolism possible [lecture ~1:11:28; 1:11:59]

★ Test cue: mosquito bite vs appendicitis = "the body mobilizes only what's useful" [lecture ~1:11:28]

Inflammatory markers — the three she expects

- Inflammatory markers = substances in the blood **in the presence of inflammation** [lecture ~36:00]

- ↑ markers = **injury is somewhere in the body** — doesn't tell you **where** or **what's causing it** [lecture ~36:00]
- **CRP (C-reactive protein)**: produced mainly in the **liver**; regulates **cytokine release** → helps regulate **inflammation** [lecture ~36:38; sheet p.17]
- **Fibrinogen**: yes, it's a **clotting factor**, but also produced in higher amounts during inflammation → helps **locate pathogens so they can be destroyed** [lecture ~36:59; sheet p.17]
- **ESR (erythrocyte sedimentation rate / "sed rate")**: spin blood in a centrifuge, time how fast RBCs drop [lecture ~37:29]
- *Textbook note*: the standard ESR lets blood settle by gravity in an upright tube for 1 hr. Fibrinogen and other proteins make RBCs clump (rouleaux), so they fall faster. On her test, use her explanation. [textbook — not in lecture]
- In inflammation, **CRP + fibrinogen stick to RBCs** → RBCs **heavier** → sediment **faster** → **ESR increases** [lecture ~38:09; sheet p.17]

★ Test cue: "Those are the only three I'm going to expect you to recognize for the test" (CRP, fibrinogen, ESR) [lecture ~39:47]

- Don't memorize each disease's lab list — just ask "**is there inflammation present?**" If yes → these markers are ↑ [lecture ~39:00]

Lab / test table

Test	What it shows	Direction	Her notes
Temperature	Inflammation (fever)	↑	Also infection [lecture ~1:05:26]
WBC count (CBC)	Leukocytosis	↑	Whole-body increase [lecture ~1:05:58]
CRP	Inflammation; liver-made	↑	Regulates cytokines [lecture ~36:38]
Fibrinogen	Inflammation	↑	Clotting factor + locates pathogens [lecture ~36:59]
ESR	Inflammation	↑	RBCs heavier → sediment faster [lecture ~38:09]
Wound culture	Infection vs inflammation	—	Plus watching for pus [lecture ~1:08:11]
Cultures / WBC count	Workup when infection suspected	—	Class diagnostic chart [class S5]
Autoantibodies (ANA, anti-DNA, anti-CCP, anti-phospholipids, RF)	Autoimmune disease (which one? not specific)	+	Slide list; only ANA explained aloud [S34; lecture ~49:56]

- **She gives no numeric values** (no reference ranges, no "CRP > X") for any lab in this lecture [lecture ~36:38-39:47]
- **Complement levels** are not used as a lab test here — complement is discussed only as the *mechanism* (complexes activate complement → inflammation) [lecture ~45:04]
- Inflammatory markers **can ↑ in all** of: immunodeficiency, hypersensitivity, autoimmunity — not specific [class S5]

★ Test cue: class practice question — the lab that **most reliably indicates infection** is a **positive urine culture**, not ESR/WBC/ANA (EWSB: ESR, WBC, ANA are all nonspecific) [class S8]

7. Telling the major immune disorders apart

The four-way comparison (class slides 3-4)

- **Big picture**: the goal is to tell apart **immunodeficiency vs autoimmunity vs hypersensitivity vs allergies** [class S3]
- **Hypersensitivity** = the umbrella term: the immune system reacts to an antigen **most people would not react to** → not normal, not appropriate, usually **excessive** → can damage host tissue [lecture ~1:14:17]

- **Allergy** = a *specific type* of hypersensitivity: immune system reacts to an **environmental antigen** that doesn't bother most people (her example: **cat dander**) [lecture ~1:14:50]
- **Autoimmunity** = immune system reacts to a **self-antigen** → also a type of hypersensitivity disorder [lecture ~1:15:23]
- So: **hypersensitivity** ⊃ {**allergy, autoimmune disease**} — she stresses people use "hypersensitivity" and "allergy" interchangeably but they are **not the same thing** [lecture ~1:14:50; class S3]

Problems each one causes

Disorder	Core problem	
Immunodeficiency	→ infection	[class S4]
Hypersensitivity	→ allergic reactions → unnecessary inflammation	[class S4]
Autoimmunity	→ damage to self tissue	[class S4]

Diagnostics

Disorder	What you check	
Immunodeficiency	Cultures, WBC count	[class S5]
Hypersensitivity	Allergy testing (e.g., skin testing)	[class S5]
Autoimmunity	Autoantibodies	[class S5]

- **All three can cause inflammatory markers to increase** — so an elevated ESR/CRP alone does not tell you which category [class S5]
- Autoimmune workup: **inflammatory markers + autoantibodies** are the two test groups [S34; lecture ~35:31]
- Inflammatory markers only say **there is injury/inflammation somewhere** — not where or why [lecture ~35:45]

Medications

Disorder	Drug approach	
Immunodeficiency	Antimicrobials	[class S6]
Hypersensitivity	Antihistamines and/or anti-inflammatory drugs	[class S6]
Autoimmunity	Anti-inflammatory drugs, immunosuppressants	[class S6]

★ Test cue: **infection is the hallmark of immunodeficiency**; a positive urine culture is the most reliable infection indicator — not ESR, not ANA [class S5, S8]

★ Test cue: the classic risk-factor question = **female with a mother who has SLE** → genetic/family + gender factors for autoimmunity [class S9; lecture ~30:00, ~31:37]

8. Hypersensitivity Types I-IV

Her memory trick — ABCD (exactly as on the class slide)

- **Type I - Allergies**
- **Type II - antiBodies**
- **Type III - immune Complex (Complement)**
- **Type IV - Delayed** [class S2]
- Second half of same slide: **Type I = IgE-mediated · Type II = antibody-mediated/cytotoxic · Type III = immune complex-mediated · Type IV = T cell-mediated/cellular** [class S2]
- Alternate mnemonic (textbook version): **ACID** = Anaphylactic/Allergic, Cytotoxic, Immune complex, Delayed [**textbook — not in lecture**]

Predisposing factors

- **Genetic predisposition + exposure to the triggering antigen** — both are needed; miss one and the disorder shouldn't develop [S62; lecture ~1:16:20]

- Exposure routes she names: **inhale** (pollen), **ingest** (peanuts), **inject** (penicillin), **skin contact** (latex, poison ivy) [lecture ~1:17:01]

The big organizing idea

- **Types I, II, III = antibody-mediated (B cells); Type IV = T cell-mediated** [lecture ~1:17:36]
- Many different syndromes live under each type; **symptoms differ wildly but the underlying pathological event is the same within a type** [lecture ~1:18:08]

Type I — Allergies (IgE-mediated)

- **Mechanism:** allergen exposure → B cells make **IgE** → IgE sticks to **mast cell / basophil membranes** → **sensitization** → next exposure binds IgE → **degranulation** → release of chemical mediators of inflammation (histamine, etc.) → symptoms [lecture ~1:19:16; ~1:21:21]
- **Sensitization:** no reaction is possible until IgE is on the mast cells/basophils; IgE may be made after the **1st, 10th, or 50th** exposure — nobody knows why [lecture ~1:20:19; ~1:22:45]
- **Timing:** reaction within **15-30 minutes** of exposure [lecture ~1:37:42]
- **Textbook note:** Type II timing varies by disorder; these are her rule-of-thumb times. [textbook — not in lecture]
- **Examples:** allergic rhinitis (hay fever), asthma, atopic dermatitis (eczema), urticaria (hives), angioedema, anaphylaxis [566, 569]
- **Why symptoms differ:** depends on **where** the mast cells/basophils degranulate — airways → asthma; nose/eyes → rhinitis; skin → eczema or hives; **system-wide** → **anaphylaxis** [lecture ~1:27:27]

Type II — antiBodies (antibody-mediated / cytotoxic)

- **Mechanism:** **IgG and IgM** made against an antigen → when the antibody **binds the antigen, that interaction damages the cell** the antigen is on [lecture ~1:30:28]
- **Not IgE** — that's the difference from Type I [lecture ~1:30:28]
- **Examples: transfusion reaction** (type A recipient has anti-B antibodies → given type B blood → donor cells destroyed); **myasthenia gravis** (antibodies block/damage acetylcholine receptors at the neuromuscular junction → muscles never get the message to contract) [lecture ~1:31:00; ~1:31:33]
- **Timing:** with Type I, reaction within **15-30 minutes** [lecture ~1:37:42]
- **Textbook note:** Type II timing varies by disorder; these are her rule-of-thumb times. [textbook — not in lecture]
- **Sensitization:** antigen exposure → IgG/IgM production [sheet p.19]

Type III — immune Complex (Complement)

- **Mechanism:** **IgG/IgM** bind antigen → form an **antigen-antibody complex (immune complex)** → complexes **deposit in tissue** → **activate complement** → **inflammation** causes the symptoms [lecture ~1:32:10; ~1:33:09]
- **Key contrast vs Type II:** in Type II the *binding itself* does the damage; in Type III the *deposited complex* → *complement* → *inflammation* does the damage [lecture ~1:33:42]
- **Example: SLE / lupus** — "a great example of a type 3 immune complex related hypersensitivity disorder" [lecture ~1:34:13]
- **Timing:** peaks around **~6 hours** after exposure to the trigger (still counted as "immediate") [lecture ~1:37:56]
- **Sensitization:** antigen exposure → IgG/IgM production → immune complex [sheet p.19]

Type IV — Delayed (T cell-mediated)

- **Mechanism:** **T cells** (not antibodies) get **sensitized** to an antigen → on re-exposure the T cells attack → that attack causes the symptoms [lecture ~1:34:44]
- **Examples:** **poison ivy / poison oak contact dermatitis** (substance in the leaves sensitizes T cells → dermatitis where you touched the plant); **type 1 diabetes** (sensitized T cells, not antibodies, attack pancreatic cells); **certain kinds of transplant rejections** [lecture ~1:35:16; ~1:35:49]

- **Timing:** peaks **1-3 days** after exposure (her example: poison ivy rash is a day or two later, not 15 minutes) [lecture ~1:38:30]
- **Sensitization:** T cells sensitized to antigen → T cells attack on re-exposure [sheet p.19]

Immediate vs delayed

- **Immediate = Types I, II, III** (antibody/B-cell) [lecture ~1:37:23]
- **Delayed = Type IV only** (T-cell) [lecture ~1:37:23]
- Timings: **I & II ≈ 15-30 min → III ≈ 6 hr peak → IV ≈ 1-3 days peak** [lecture ~1:37:42]
- **Textbook note:** Type II timing varies by disorder; these are her rule-of-thumb times. [textbook — not in lecture]
- Sheet shorthand: **Type I, II, III: Immediate/B-cell · Type IV: delayed/T-cell** [sheet p.19]

★ Test cue: **anything described as an "allergy" = Type I, IgE-mediated** — that's the one exception she'll ask you to identify from a symptom description [lecture ~1:36:52]

★ Test cue: she **cannot** ask you to tell Type II from Type IV by symptoms — you can't tell from observation which was done by an antibody vs a sensitized T cell [lecture ~1:36:11]

9. Allergy and anaphylaxis

Core definitions and mechanism

- **Allergen** = the antigen in a Type I reaction that causes the allergy (she calls Type I antigens "allergens" because they cause allergies) [lecture ~1:19:44]
- **Allergy** = hypersensitivity to an **environmental antigen** that doesn't bother most people (cat dander example) [lecture ~1:14:50]
- **Process:** exposed to allergen (pollen, cat dander, peanuts) → B cells make **IgE** → IgE sticks to **mast cell + basophil membranes** → person is **sensitized** [lecture ~1:19:44; ~1:21:21]
- **Mast cells / basophils** hold granules full of chemical mediators of inflammation (histamine, prostaglandins, leukotrienes, bradykinin, cytokines) [lecture ~56:48; ~57:22]
- **Degranulation** = allergen binds the IgE on mast cells/basophils → they dump their chemical mediators → **those mediators cause the symptoms** [lecture ~1:21:21; sheet p.19]
- **Sensitization** = having IgE antibodies on the mast cell/basophil membranes; reactions only happen after this [lecture ~1:21:21]
- No allergic reaction is possible **until sensitization occurs** [lecture ~1:22:45]

Nursing safety point she emphasizes

- **We can never stop monitoring patients for allergic reactions to medications** — IgE might start being made after the 1st, 10th, or 50th exposure; her example: patients fine through **two full courses of penicillin**, then reactive forever after [lecture ~1:23:32]
- Help patients **identify** allergens (skin testing: little bits of antigen under the skin → see which cause a reaction) so they can **avoid** them [lecture ~1:29:22]

Type I examples and their symptoms

Disorder	Her symptom description	
Allergic rhinitis (hay fever)	runny nose, sneezing, red/itchy/watery eyes	[lecture ~1:24:49]
Asthma	airway constriction, wheezing, shortness of breath	[lecture ~1:25:12]
Atopic dermatitis (eczema)	itchy, scaly, crusty rashes — "incredibly itchy"	[lecture ~1:25:12]
Urticaria (hives)	looks like mosquito bites all over; very itchy	[lecture ~1:25:43]
Angioedema	tongue, lips, part of the face start to swell	[lecture ~1:25:43]

Disorder	Her symptom description
Anaphylaxis	most severe Type I; see below [lecture ~1:26:14]

- Full example list also on the recorded deck: allergic rhinitis, asthma, atopic dermatitis, urticaria, angioedema, anaphylaxis [S66, S69]
- Same underlying event (IgE → degranulation) in all of them; symptoms differ by **where degranulation happens** [lecture ~1:27:15]

Anaphylaxis

- **Most severe** Type I disorder [lecture ~1:26:14]
- **Signs: blood pressure drops → hypotension is the primary worry**; also bronchoconstriction, some shortness of breath; if BP drops enough → **shock (anaphylactic shock)**; **people can die** [lecture ~1:26:14]
- **Mechanism of the drop:** degranulation **system-wide** → whole-body **vasodilation** from the inflammatory response → BP drops → anaphylaxis [lecture ~1:27:48; sheet p.19]
- **Treatment (her words):** she gives **no** anaphylaxis drug/nursing steps in this lecture — only that these are Type I disorders treated with antihistamines/anti-inflammatories and that we never stop monitoring [lecture ~1:29:05; S70]
- **Epinephrine as first-line for anaphylaxis** [textbook — not in lecture]

Interventions for Type I / allergy

- **Avoid the trigger** (we don't know how to desensitize once sensitized) + **treat the symptoms** [lecture ~1:39:06]
- *Textbook note:* allergy shots (immunotherapy) and supervised drug desensitization do exist. [textbook — not in lecture]
- **Medications: antihistamines** (histamine is one of the mediators) + **anti-inflammatory drugs** [S70; lecture ~1:29:05]
- **Corticosteroids — local vs systemic** (example: **fluticasone = Flonase/Flovent**) [S70]
- Some triggers are hard to avoid (common trees, pets) → then we lean more on medications [lecture ~1:29:42]

★ Test cue: anaphylaxis = **systemic degranulation → vasodilation → hypotension → shock**; the drop in BP is the thing she says to worry about most
[lecture ~1:26:14; ~1:27:48]

★ Test cue: "never stop monitoring for drug allergy," even after prior uneventful courses [lecture ~1:22:45]

10. Transplant rejection

- **Tissue typing for transplants** looks at **MHC molecule genes** — "is this person's tissue similar enough to that patient's so that the patient will not reject that donated kidney" [lecture ~19:08]
- Because MHC genes are so variable (**128 genes, 50-650 alleles each**), two people are almost never a genotype match **unless they're identical twins** [lecture ~16:26]
- *Textbook note:* full siblings have about a 25% chance of being an HLA match. [textbook — not in lecture]
- The old term for MHC in this kind of DNA testing is **HLA (human leukocyte antigen)** — you still hear "HLA testing" in the news [lecture ~20:41]
- **Certain kinds of transplant rejections are Type IV (T cell-mediated) reactions** [lecture ~1:35:49]
- **Hyperacute rejection** = preformed antibodies against donor antigens attack the graft within minutes to hours [textbook — not in lecture]
- **Acute rejection** = T-cell and/or antibody response days to months after transplant [textbook — not in lecture]
- **Chronic rejection** = gradual loss of graft function over months to years, with vascular fibrosis and scarring [textbook — not in lecture]

11. Quick-review tables

Table 1: Hypersensitivity Types I-IV

Type	Mediator	Timing	Her examples	Source
Type I – Allergies / IgE-mediated	IgE (B cells make it); sticks to mast cells & basophils → degranulation	Immediate , ~ 15-30 min after exposure	Allergic rhinitis (hay fever), asthma, atopic dermatitis (eczema), urticaria (hives), angioedema, anaphylaxis	[class S2; S66, S69; lecture ~1:37:56]
Type II – antiBodies / antibody-mediated (cytotoxic)	IgG & IgM bind antigen → damage the cell the antigen is on	Immediate , ~ 15-30 min	Transfusion reaction (type A recipient given type B blood); myasthenia gravis (anti-acetylcholine receptor)	[class S2; lecture ~1:30:28, ~1:37:56]
Type III – immune Complex	IgG & IgM → antigen-antibody (immune) complexes → deposit in tissue → complement → inflammation	Immediate category, but peaks ~ 6 hours	SLE / lupus	[class S2; lecture ~1:32:07, ~1:33:09, ~1:37:56]
Type IV – Delayed	T cells (no antibodies)	Delayed , peaks 1-3 days	Poison ivy/poison oak contact dermatitis; type 1 diabetes; certain transplant rejections	[class S2; lecture ~1:34:44, ~1:35:49, ~1:38:30]

★ Test cue: **ABCD memory trick** — Type I Allergies, Type II antiBodies, Type III immune Complex (Complement), Type IV Delayed [class S2].

★ Test cue: If it "sounds like an allergy" → it's **Type I, IgE-mediated** [lecture ~1:36:52].

Table 2: Autoimmune diagnostic tests

Test	What it detects	Used for	Source
CRP (C-reactive protein)	Protein made mostly in the liver ; regulates cytokine release → regulates inflammation	Inflammation marker	[S31, S34; lecture ~36:33]
Fibrinogen	Clotting factor, also ↑ in inflammation; helps locate pathogens so they can be destroyed	Inflammation marker	[S32; lecture ~37:04]
ESR (erythrocyte sedimentation rate)	Rate RBCs sediment in a centrifuge; ↑ because markers stick to RBCs → heavier → drop faster	Inflammation marker	[S33; lecture ~37:37, ~38:09]
ANA (anti-nuclear antibody)	Antibodies attacking nuclear/DNA antigens	Autoimmune (primary autoantibody in lupus)	[S34; lecture ~39:49, ~43:32]

Test	What it detects	Used for	Source
Anti-DNA	Slide-listed autoantibody only	Autoimmune work-up (listed)	[S34 — slide-only, not named in lecture]
Anti-CCP	Slide-listed autoantibody only	Autoimmune work-up (listed)	[S34 — slide-only, not named in lecture]
Anti-phospholipids	Slide-listed autoantibody only	Autoimmune work-up (listed)	[S34 — slide-only, not named in lecture]
Rheumatoid factor (RF)	Slide-listed autoantibody only	Autoimmune work-up (listed)	[S34 — slide-only, not named in lecture]

What the slide-only antibodies usually point to (she didn't explain these; background only):

- **Anti-DNA (anti-dsDNA)** → fairly specific for **SLE**; often tracks lupus kidney disease **[textbook — not in lecture]**
- **Anti-CCP** → highly specific for **rheumatoid arthritis** **[textbook — not in lecture]**
- **Antiphospholipid antibodies** → ↑ risk of **blood clots and pregnancy loss**; seen in SLE and antiphospholipid syndrome **[textbook — not in lecture]**
- **RF (rheumatoid factor)** → common in **RA**, but less specific than anti-CCP (also in other diseases and some healthy older adults) **[textbook — not in lecture]**

★ Test cue: **Two groups of tests** = inflammatory markers + autoantibodies [S30; lecture ~35:28].

★ Test cue: Markers tell you **inflammation is present, not where or why**; autoantibodies tell you **probably autoimmune, not which disease** [lecture ~36:00, ~49:55].

Table 3: Inflammation labs

Lab	Rises / means	Source
CRP	↑ with inflammation/injury	[S31; lecture ~36:33]
Fibrinogen	↑ with inflammation	[S32; lecture ~37:04]
ESR	↑ with inflammation (RBCs heavier, sediment faster)	[S33; lecture ~38:09]
WBC count	↑ = leukocytosis (part of acute-phase response)	[S56; lecture ~1:05:26]
Temperature	Fever = part of acute-phase response	[S55; lecture ~1:05:26]

★ Test cue: Only these **three** inflammatory markers "I'm going to expect you to recognize for the test" [lecture ~39:49].

★ Test cue: Fever + ↑WBC ≠ infection by itself — inflammation from surgery or MI does it too [class S8; lecture ~1:05:58].

Table 4: Exudate types

Type	Look / meaning	Source
Serous	Clear, light yellowish — serum without cells	[S53; lecture ~1:01:08]
Serosanguineous	"Watery blood," pink (sero = serous + sanguineous = blood)	[S53; lecture ~1:01:40]
Purulent	Pus — infected-looking, dead WBCs in it	[S53; lecture ~1:01:40]

- Exudate = fluid collecting in a wound from ↑ **capillary permeability**; usually a *good* thing — brings nutrients, O₂, antibodies [lecture ~1:01:08, ~1:02:13; sheet p.18]

Table 5: Immunodeficiency vs autoimmunity vs hypersensitivity vs allergy

	Problem	Diagnostics	Meds	Source
Immunodeficiency	→ infection	Cultures, WBC count	Antimicrobials	[class S4, S5, S6]
Autoimmunity	→ damage to self tissue	Check for autoantibodies	Anti-inflammatory drugs, immunosuppressants	[class S4, S5, S6]

	Problem	Diagnostics	Meds	Source
Hypersensitivity (broad umbrella)	Immune system reacts to an antigen most people don't react to ; excessive → unnecessary inflammation	Allergy testing	Antihistamines and/or anti-inflammatory drugs	[class S4-S6; lecture ~1:14
Allergy	A specific type of hypersensitivity → reacts to an environmental antigen (cat dander, pollen)	Allergy testing (skin testing)	Antihistamines, anti-inflammatories, corticosteroids	[lecture ~1:14:50; S70]

- **All can cause inflammatory markers to increase** [class S5, S6].
- Allergy AND autoimmune disease are both **specific examples** under the hypersensitivity umbrella [lecture ~1:15:23, ~1:15:54].

12. Likely test traps

- **Inflammation ≠ infection.** Inflammation is the tissue's response to **injury**; infection requires a **pathogen** [lecture ~53:07, ~54:09].
- **"-itis" means inflammation, not infection.** Appendicitis/pericarditis ≠ automatically infected [lecture ~54:41].
- **ESR/CRP/fibrinogen are non-specific.** They say "injury is somewhere" — not where, not why [lecture ~36:00].
- **Autoantibodies don't name the disease.** Positive = "probably autoimmune," but not which one [lecture ~49:55].
- **No single test confirms lupus.** Diagnosis of exclusion — "the great imitator" [lecture ~48:53, ~49:24].
- **Fever + ↑WBC ≠ infection.** Post-op incision or MI = injured tissue → inflammation [lecture ~1:05:58, ~1:07:37, ~1:09:45].
- **Watch trends,** not single values: infection → keeps climbing; inflammation → plateaus then drops [lecture ~1:08:11].
- **Type II vs Type III, both IgG/IgM:** Type II = the **binding itself** damages the cell; Type III = complexes **deposit** → complement → **inflammation** causes symptoms [lecture ~1:33:09, ~1:33:42].
- **Lupus = Type III,** not Type II or IV [lecture ~1:34:13; sheet p.19].
- **Type 1 diabetes appears twice** — organ-specific autoimmune disease AND a Type IV (T-cell) hypersensitivity example [lecture ~33:50, ~1:35:49].
- **IgE = Type I/allergy only.** Types II & III use IgG/IgM [lecture ~1:19:11, ~1:30:28, ~1:32:38].
- **"Allergy" ⇒ Type I, IgE-mediated** — the one type she expects you to ID from symptoms [lecture ~1:36:52].
- **No reaction until sensitized.** IgE must be stuck on mast cells/basophils first [lecture ~1:23:00].
- **Sensitization timing is unpredictable** — 1st, 10th, or 50th exposure; **never stop monitoring for drug allergy** [lecture ~1:23:32].
- **Symptoms differ by WHERE mast cells degranulate** — same mechanism, different syndrome (airway = asthma, skin = hives, system-wide = anaphylaxis) [lecture ~1:27:15].
- **Anaphylaxis: the big worry is hypotension → shock** (whole-body vasodilation), plus bronchoconstriction [lecture ~1:26:14, ~1:28:21].
- **Self-tolerance wording:** goal is the immune system **tolerates self-cells**; failure = autoimmune disease [lecture ~1:04, ~1:37].
- **Three self-tolerance mechanisms:** regulatory T cells, bone marrow/thymus screening, and MHC display requirement [lecture ~2:13, ~2:48, ~3:20].
- **MHC is "the most important way"** you have self-tolerance [lecture ~3:20; S7].

- **T cell won't activate unless its antigen is on the RIGHT MHC class** — protective mechanism against accidental activation/autoimmunity [class S10; lecture ~8:12, ~11:29].
- **Class I = all nucleated cells + cytotoxic T (CD8); Class II = APCs + helper T (CD4)** — $CD4 \times 2 = 8$, $CD8 \times 1 = 8$ [S9; lecture ~14:44].
- **Class I shows intracellular antigens** (self, cancer, viral); **Class II shows anything phagocytized** [lecture ~9:49, ~6:34].
- **Hypersensitivity needs BOTH** genetic predisposition AND exposure to the trigger — one alone isn't enough [lecture ~1:16:28].
- **Types I, II, III = immediate (B-cell/antibody); Type IV = delayed (T-cell)** — timings 15–30 min, 15–30 min, ~6 h, 1–3 days [sheet p.19; lecture ~1:37:23].
- *Textbook note:* Type II timing varies by disorder; these are her rule-of-thumb times. [textbook — not in lecture]
- **Antigenic mimicry:** group A beta-hemolytic strep → rheumatic heart disease or glomerulonephritis → hence throat culture + antibiotics [class S11; lecture ~25:39, ~26:43].
- *Textbook note:* rheumatic heart disease is the classic mimicry example; post-strep glomerulonephritis is usually immune-complex (Type III). [textbook — not in lecture]
- **"Sequestered" = hidden/isolated.** Released antigens the immune system never learned = attacked as foreign [lecture ~27:52, ~28:25].
- **Remission vs flare:** worse = exacerbation/flare; better = remission; goal = keep them in remission [S28; lecture ~32:40].
- **Corticosteroids** = powerful anti-inflammatory = **synthetic cortisol**, lots of side effects; **NSAIDs** = fewer side effects but weaker [lecture ~51:28, ~51:58].
- **Exudate is usually GOOD** — fluid, nutrients, O_2 , antibodies, wound cleaning [lecture ~1:02:13].
- **Lupus flares are triggered by sun exposure** — keep patients out of the sun [lecture ~47:19].

13. Vocabulary list, defined

- **Allergen** — The antigen in a Type I reaction; "we sometimes call them allergens because they're the antigens that cause the allergies" (e.g., pollen, cat dander, peanuts) [lecture ~1:19:44].
- **Allergic rhinitis/hay fever** — Type I example: runny nose, lots of sneezing, red itchy watery eyes [S66; lecture ~1:25:12].
- **Anaphylaxis** — "Probably the most severe of the type 1 disorders"; degranulation system-wide → BP drops (hypotension) → can go into anaphylactic shock; people can die; may also have bronchoconstriction and SOB [lecture ~1:26:14, ~1:27:48; sheet p.19].
- **Angioedema** — Type I example: tongue, lips, and part of the face start to swell [S66; lecture ~1:25:43; sheet p.19].
- **Anti-DNA** — Slide-listed autoantibody on the autoimmune diagnostic test list. ⚠️ Slide-only — she never names or defines it in the recording (she describes lupus antibodies "against the antigens in that DNA" but calls them ANA) [S34].
- **Antigen-antibody complex/immune complex** — "An immune complex is just a fancy word for the antigen-antibody complex... When the antigen and the antibody bind together, that's an immune complex" [lecture ~1:32:07].
- **Antinuclear antibody/ANA** — "Antibodies that attack antigens that are in the nucleus"; a primary autoantibody in lupus; forms after damaged cells expose DNA [lecture ~43:32, ~44:02].
- **Asthma** — Type I example: airway constriction, wheezing, shortness of breath; from mast cell/basophil degranulation in the airways [lecture ~1:25:12, ~1:27:15].
- **Atopic dermatitis/eczema** — Type I example: itchy, scaly, crusty rashes — "incredibly itchy"; degranulation in the skin [lecture ~1:25:12].
- **Autoantibody** — "Auto means self... an autoantibody is an antibody that is attacking self-antigens"; expected in autoimmune disease [lecture ~39:49].
- **Autoimmunity** — The immune system attacking self-cells/self-antigens; those self cells get injured → organ or body system doesn't work properly [lecture ~22:17].

- **Butterfly rash/malar rash** — Classic lupus skin symptom; butterfly shape — wings on the cheeks, body over the bridge of the nose; worsened by sun exposure [S40; lecture ~46:46, ~47:19].
- **C-reactive protein/CRP** — Inflammatory marker produced primarily in the **liver**; regulates release of cytokines and thus helps regulate inflammation; ↑ in inflammation [lecture ~36:33; sheet p.17].
- **Catabolism** — "Breaking down protein"; skeletal muscle catabolism = breaking down skeletal muscle to get amino acids for antibodies or tissue repair [lecture ~1:10:21, ~1:10:52].
- **Degranulation** — Mast cells and basophils releasing their granules full of chemical mediators of inflammation (histamine, prostaglandins, leukotrienes, bradykinin, cytokines) [lecture ~56:48, ~57:22, ~1:21:52].
- **Delayed hypersensitivity disorder — Type IV only**; T-cell-mediated; peaks **1-3 days** after exposure (e.g., poison ivy rash a day or two later) [lecture ~1:37:23, ~1:38:30].
- **Erythrocyte sedimentation rate/ESR** — Timing how fast RBCs sediment to the bottom of a spun specimen; ↑ in inflammation because CRP/fibrinogen stick to RBCs → heavier → drop faster [lecture ~37:37, ~38:09; sheet p.17].
- **Exudate** — "A lot of fluid collecting in that wound" from ↑ capillary permeability; types: serous, serosanguineous, purulent [lecture ~1:00:33, ~1:01:08; S53].
- **Fibrinogen** — A clotting factor, also produced in higher amounts during inflammation; helps locate pathogens so they can be destroyed [lecture ~37:04; sheet p.17].
- **Hypersensitivity** — The broad umbrella term: immune system reacts to an antigen most people would not react to; not normal, not appropriate, usually excessive → can damage host tissue [lecture ~1:14:17, ~1:15:54].
- **Immediate hypersensitivity disorder** — Types **I, II, III** (B-cell/antibody); Type I & II react ~15-30 min after exposure, Type III peaks ~6 hours [lecture ~1:37:23, ~1:37:56; sheet p.19].
- **Textbook note:** Type II timing varies by disorder; these are her rule-of-thumb times. **[textbook — not in lecture]**
- **Inflammation** — "The tissue's response to injury" [S45; lecture ~53:07].
- **Major histocompatibility complex/MHC (Class I and Class II)** — "Self-recognition proteins" on nearly every cell; display antigens for T cells to inspect. **Class II** = on APCs/phagocytes → works with **helper T (CD4)**; **Class I** = all nucleated cells → displays intracellular antigens (self, viral, cancer) → works with **cytotoxic T (CD8)** [S7, S9; lecture ~3:51, ~5:33, ~10:55].
- **Mast cells** — "Basically just a basophil that has moved into the tissues"; granules hold chemical mediators of inflammation; IgE sticks to their membranes in Type I [lecture ~57:07, ~1:20:49].
- **Textbook note:** mast cells and basophils are related but separate cell lines; basophils don't turn into mast cells. **[textbook — not in lecture]**
- **Plasmapheresis** — A procedure used "to try and stop the disease process where the immune system is attacking self" (an intervention for autoimmune disease) [lecture ~41:58]. ⚠ Not defined further in the Immune 2 sheet [sheet p.10 — outside section].
- **Self-reactive** — Cells "that might go out and attack self cells"; regulatory T cells suppress them, bone marrow/thymus screen them out [lecture ~2:13, ~2:48].
- **Self-tolerance** — "We want our immune system to tolerate our self-cells. We don't want there to be an immune response directed at our own cells and our own antigens" [lecture ~1:04].
- **Sensitization** — When IgE antibodies stick to the cell membranes of mast cells and basophils — "that person is now considered sensitized to that allergen"; no allergic reaction is possible until this happens [lecture ~1:21:21, ~1:23:00].
- **Sequestered** — "To hide away or to isolate" (like a sequestered jury); sequestered self-antigens are hidden from the immune system during fetal life, so it never learns they're self [lecture ~27:52, ~28:25].
- **Systemic lupus erythematosus/SLE** — Systemic autoimmune disease; ANA and other autoantibodies → antigen-antibody complexes deposit in collagen-rich tissue → complement → inflammation; malar rash, arthritis, renal failure, multi-organ; "the

great imitator," diagnosis of exclusion

[S36; lecture ~43:12, ~44:33, ~47:50, ~48:53; sheet p.17].

- **Type I, IgE-mediated disorder** — B cells make **IgE** against an allergen → IgE sticks to mast cells/basophils → re-exposure → degranulation → chemical mediators → symptoms; ~15-30 min [class S2; lecture ~1:19:11, ~1:21:21].
- *Textbook note:* Type II timing varies by disorder; these are her rule-of-thumb times.
[textbook — not in lecture]
- **Type II, antibody-mediated/cytotoxic disorder** — **IgG/IgM** bind antigen → that binding damages the cell the antigen is on (e.g., transfusion reaction, myasthenia gravis) [class S2; lecture ~1:30:28, ~1:33:09].
- **Type III, immune complex-mediated disorder** — **IgG/IgM** form antigen-antibody complexes → deposit in tissue → activate **complement** → **inflammation** = symptoms (e.g., lupus) [class S2; lecture ~1:32:07, ~1:33:42].
- **Type IV, T cell-mediated disorder** — Sensitized **T cells** attack the antigen; no antibodies; **delayed** (1-3 days); e.g., poison ivy dermatitis, type 1 diabetes, certain transplant rejections [class S2; lecture ~1:34:44].
- **Urticaria/hives** — Type I example: looks like mosquito bites all over the body, very itchy; degranulation in the skin [lecture ~1:25:43; sheet p.19].

Class 5 — Immunity & Inflammation, Part 2: Autoimmunity & Hypersensitivity

NRSRG 224 Pathophysiology · Fall 2026 · filled from the class study guide

Personal study sheet — not for submission.

Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. **not covered in lecture** = textbook knowledge, not from her lecture.

Sources: recorded lecture subtitles (times are the 1x recording, about ± 30 s; on the 2x video, halve them), recorded and class slides, verified clinical excerpts, and your full study sheet pp. 17-19.

Page 1 — Distinguishing self from nonself

Guide: "Self-tolerance (the absence of an immune response directed against a person's own antigens) normally is maintained through:" — two blank dashes, then MHC molecules (3rd mechanism, already printed).

A1 Self-tolerance mechanism #1 (blank -)

- Mechanism #1 = regulatory T cells (aka suppressor T cells) [02:13]
- Two jobs: shut off immune response once pathogen is gone; suppress any potentially self-reactive WBCs [02:13]
- Gets rid of those self-reactive cells before they can engage in autoimmune activity [02:48]
- Sheet version: turn off immune response + suppress self-reactive WBCs [sheet p.17]

A2 Self-tolerance mechanism #2 (blank -)

- Mechanism #2 = bone marrow + thymus screening of immature WBCs [02:48]
- T cells mature in thymus, B cells mature in bone marrow; both sites screen for potentially self-reactive cells [02:48]
- If found → get rid of self-reactive cells BEFORE release into general circulation [03:20]
- Sheet version: screen + remove immature self-reactive B cells and T cells [sheet p.17]

Page 1 — Inflammation

B1 S/S (cardinal signs): list all 5 (fills the five blank dashes) and for each, the mechanism she gave

- Redness + heat ← vasodilation → more blood to area [58:27]
- Swelling (edema) ← ↑ capillary permeability → fluid leaks into tissues [59:31]
- Pain ← swelling; usually goes along with it [59:31]
- Impaired/loss of function ← swollen parts don't work (e.g., sprained finger can't bend) [1:00:02]
- Cardinal signs listed: redness, heat, swelling, pain, impaired function [sheet p.18]

B2 Vascular response: momentary vasoconstriction, then vasodilation → ____

- Momentary vasoconstriction → vasodilation [sheet p.18]
- Vasodilation = first vascular event → brings more blood: O₂, fuel, antibodies to fight

infection [58:27]

- Result = the redness and warmth of inflammation [58:59]

B3 ↑ capillary permeability → fluid escapes into tissues → ____

- ↑ capillary permeability = capillaries get 'leaky' → fluid leaks into tissues, bringing nutrients + antibodies [59:31]
- Causes edema, pain, and loss of function [59:31]
- Exudate types: serous (clear, light yellow = serum), serosanguineous (watery blood/pink), purulent (pus = dead WBCs, infected) [1:01:08]; [slide 53]
- Exudate is usually GOOD: brings nutrients/O₂/antibodies, flushes dirt, dilutes irritants, slows spread of contaminants [1:02:13]

B4 Cellular response - movement of WBCs into area of injury → ____

- Cellular response = phagocytic WBCs migrate to the injury area [1:03:53]
- Neutrophils get there first, followed by monocyte-macrophages [1:03:53]
- They phagocytize pathogens + dead tissue → get it ready for repair [1:03:53]
- If needed, they act as APCs → signal/activate adaptive immunity [1:04:24]

B5 Acute-phase response (fever, ↑ # WBCs, skeletal muscle catabolism) — what to know

- Acute-phase response = systemic effects: fever, leukocytosis, ↑ inflammatory markers, varying degrees of skeletal muscle catabolism [slide 57]; [1:05:39]
- Fever is beneficial: bacteria can't survive higher temp + WBCs become more active [1:05:26]
- Leukocytosis = actual ↑ in number of WBCs; fever + ↑ WBC ≠ always infection (post-op incision, MI) → monitor trends, check pus, wound culture [1:05:39]
- Skeletal muscle catabolism = break down muscle protein → amino acids to make antibodies / repair tissue; body mobilizes only what's useful (mosquito bite vs appendicitis) [1:10:21]
- Markers: CRP (liver; regulates cytokines), fibrinogen (helps locate pathogens), ESR (markers stick to RBCs → heavier → sediment faster) [36:36]; [37:37]
- ★ **Prof emphasis:** 'those are the only three I'm going to expect you to recognize for the test' (CRP, fibrinogen, ESR) [39:49]

B6 Potential problems r/t inflammation

- Inflammation can get out of control: very severe, or recurrent/persistent → harmful [1:12:01]
- Chronic irritation/inflammation → metaplasia + scar tissue formation [1:12:34]

- Scar tissue → loss of function + deformity (RA joints pulled into abnormal positions) [1:13:09]
- Kidney scarring → can't filter blood / make urine like real kidney tissue [1:13:41]

B7 Potential problems if immune system function impaired/altered

- Immunodeficiency → infection [class slide 4]
- Hypersensitivity → allergic reactions → unnecessary inflammation [class slide 4]
- Autoimmunity → damage to self tissue [class slide 4]
- Diagnostics: immunodeficiency → cultures, WBC count; hypersensitivity → allergy testing; autoimmunity → check for autoantibodies [class slide 5]
- ★ **Prof emphasis:** ALL three can cause inflammatory markers to increase [class slide 5]

Flowchart 1 — Autoimmunity (breakdown in self-tolerance)

C1 Nursing problem:

- Nursing problem: damage to self tissue — immune system attacks self-antigens [class slide 4]
- Injured self cells → organ/body system not working properly [22:17]
- Most autoimmune diseases are chronic: cycles of remissions and exacerbations/flare-ups [32:40]
- Goal of care: minimize exacerbations, get patients into remission, keep them there as long as possible [33:14]

C2 SLE — Assessment findings:

- Ag-Ab complexes deposit in collagen-rich tissues → many body systems; skin: bruising, rashes, classic butterfly/malar rash (worsened by sun; flares triggered by sun) [46:46]; [47:19]
- Joints (lupus arthritis), kidney (inflammation, renal failure), lungs, heart, eyes, GI tract; not all systems in every patient [48:01]
- Autoantibodies: ANA (anti-nuclear) is a primary one — antibodies vs nuclear/DNA antigens; also autoantibodies vs RBCs and platelets [43:32]; [45:34]
- Labs: ↑ inflammatory markers (CRP, fibrinogen, ESR) = inflammation only; autoantibody list on slide also Anti-DNA, Anti-CCP, Anti-phospholipids, RF [slide 34]; [35:28]
- Dx is a diagnosis of exclusion — 'the great imitator'; no one definitive test; can take years [48:43]

C3 Interventions (autoimmune / SLE):

- Two levels of intervention: stop the underlying autoimmune attack + treat the symptoms [41:39]; [slide 35]
- Stop attack: immunomodulators — immunosuppressive drugs, monoclonal antibodies, corticosteroids; also plasmapheresis [41:58]; [slide 44]
- Treat symptoms/inflammation: NSAIDs (Motrin, ibuprofen, aspirin) — fewer side effects, less powerful; corticosteroids = synthetic cortisol, powerful but many side effects [51:28]

- Use NSAIDs first, add corticosteroids if necessary (NSAIDs won't halt kidney damage) [52:31]
- Keep lupus patients out of the sun — sun exposure triggers exacerbations [47:19]

Flowchart 2 — Hypersensitivity disorders

D1 Nursing problem:

- Problem: immune system reacts to an antigen most people would not react to; not normal/appropriate, usually excessive → can damage host tissue [1:14:17]
- Results in unnecessary inflammation / allergic reactions [class slide 4]
- Hypersensitivity = broad umbrella; allergy (reacts to environmental antigen, e.g., cat dander) and autoimmune disease (reacts to self-antigen) are two specific types [1:14:50]
- Predisposing factors: BOTH genetic predisposition + exposure to triggering antigen (inhaled, ingested, injected, skin contact) [1:16:20]

D2 Type I — allergic rhinitis (hay fever)

- Type I = IgE-mediated; allergic rhinitis = 'hay fever' [1:24:49]
- Findings: runny nose, lots of sneezing, red/itchy/watery eyes [1:24:49]
- Mechanism: mast cells/basophils degranulate in nasal passages / eye mucous membranes [1:27:27]

D3 Type I — asthma

- Type I, IgE-mediated; airway constriction [1:25:12]
- Findings: wheezing, shortness of breath [1:25:12]
- Mechanism: degranulation in the airways [1:27:27]
- Reaction within ~15 min of trigger exposure (e.g., cat-allergic person starts to wheeze in ~15 min) [1:37:56]

D4 Type I — urticaria

- Urticaria = hives; Type I, IgE-mediated [1:25:43]
- Looks like mosquito bites all over the body; very itchy [1:25:43]
- Mechanism: degranulation in the skin [1:27:27]

D5 Type I — atopic dermatitis (eczema)

- Atopic dermatitis = eczema; Type I, IgE-mediated [1:25:12]
- Findings: itchy, scaly, crusty rashes — 'incredibly itchy' [1:25:12]
- Mechanism: degranulation in the skin [1:27:27]

D6 Type I — anaphylaxis

- Most severe Type I disorder; main worry = blood pressure drops → hypotension → can progress to anaphylactic shock; people can die [1:26:14]
- Also bronchoconstriction and some shortness of breath [1:26:14]
- Mechanism: degranulation system-wide → whole-body vasodilation → BP drop [1:27:48]

- Related: angioedema = tongue, lips, part of face swell [1:25:43]
- Textbook, not from her lecture: anaphylaxis emergency care = IM epinephrine first, then airway/oxygen, IV fluids for hypotension, plus antihistamines/corticosteroids

[not covered in lecture — textbook]

not covered in lecture

D7 Type II — blood transfusion reactions

- Type II = antibody-mediated/cytotoxic; uses IgG and IgM (not IgE) [1:30:28]
- Example: type A recipient (has antibodies vs B antigen) given type B blood → recipient antibodies attack the donor cells [1:31:00]
- When antibody binds antigen on donor cell → destroys the donated cell; the binding itself causes the damage [1:31:00]

D8 Type III — SLE (why it is Type III)

- Type III = immune complex-mediated (IgG/IgM); immune complex = antigen-antibody complex [1:32:10]
- Not the binding itself — complexes get deposited in tissue → activate complement → inflammation = the symptoms [1:33:09]
- That is exactly the lupus scenario (ANA binds DNA antigen → complexes deposit in collagen-rich tissue) [44:33]; [1:34:13]

D9 Type IV — contact dermatitis (poison ivy), diabetes mellitus

- Type IV = T-cell-mediated / cellular — only type NOT antibody-mediated; delayed [1:34:44]

- T cells get sensitized to antigen → next exposure, T cells attack → symptoms [1:34:44]
- Examples: poison ivy/poison oak contact dermatitis (rash where touched); type 1 diabetes (sensitized T cells attack pancreatic cells); certain transplant rejections [1:35:16]
- Delayed timing: rash/peak ~1-3 days after exposure (vs Type I/II 15-30 min, Type III ~6 hr) [1:37:56]

D10 Assessment findings (hypersensitivity) — guide says 'Specific to individual disorders'

- Findings are specific to individual disorders; symptoms differ by WHERE mast cells/basophils degranulate (airways→asthma, nose/eyes→rhinitis, skin→eczema/hives, system-wide→anaphylaxis) [1:27:15]
- Diagnostics: allergy testing — skin testing, sticking little bits of antigen under the skin to see which cause a reaction [1:29:22]; [class slide 5]
- All three major immune problems can increase inflammatory markers [class slide 5]
- ★ **Prof emphasis:** on the test she can't ask Type II vs Type IV from symptoms; but anything that sounds like an 'allergy' = Type I IgE-mediated [1:36:33]

D11 Interventions (hypersensitivity):

- Main interventions: get rid of/avoid the trigger + treat the symptoms; we can't desensitize once a person is sensitized [1:39:01]
- Meds for Type I: antihistamines (histamine is a mediator) + anti-inflammatories —

corticosteroids local vs systemic (e.g., fluticasone), others [1:29:05]; [slide 70]

- Help patients identify allergens (skin testing) and avoid them; if unavoidable (common trees, pets) → rely more on medications [1:29:53]
- ★ **Prof emphasis:** we can NEVER stop monitoring patients for allergic reactions to meds — B cells may start making IgE after the 1st, 10th, or 50th exposure (penicillin story)

[1:23:32]